After Simon, after Hrushovski

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ABSTRACT. This is a rewriting of [Si] and [Hr]. But I took a few liberties in the setting and the notation.

E.g. I use a more compact notation when defining the patterns language.

I work with realizations of types over M instead of types themselves, and I use the notation $\mathcal{N}$ for a suitable set of realizations of the types over M.

I restrict the formulas used to define the pattern language because I would like to accomnodate the example in Section 5 – this only partially succeeds.

I replace the formulas $\alpha(z)$ with types (and coheirs with strong coheirs) as presented in appendix A of [Hr].

1. Patterns

In the following $\mathcal{U}$ is some monster model of a complete theory T of language L. For simplicity L is assumed to be one-sorted.

We define a new structure $\mathcal{N}$ whose domain is a subset of $\mathcal{U}^x$, where x is some fixed tuple of variables (as a matter of fact, we could safely assume $|x| = 1$). The language of $\mathcal{N}$, which is denoted by L^{P}, is defined below. First we need some notation.

Let $z = \langle z_i : i < \lambda \rangle$ be a tuple of (so-called) parameter variables. We will work with a fixed set of formulas $L_z \subseteq \Delta(x;z) \subseteq L_{x,z}$. Let V be some large set of variables that are copies of x. Finally, let Δ be the set of Boolean combinations of formulas in $\bigcup_{x' \in V} \Delta(x';z)$. The choice of Δ is justified in Remark 1.

Let $A \subseteq \mathcal{U}^x$ and $B \subseteq \mathcal{U}^z$. We write A, Δ for the set of formulas $\varphi(\bar{a};z)$ for $\varphi(\bar{x};z) \in \Delta$ and $\bar{a} \in A^{\bar{x}}$, and we write Δ, B for the set of formulas $\varphi(\bar{x};b)$ for $\varphi(\bar{x};z) \in \Delta$ and $b \in B^z$.

Let $\bar{a}, \bar{a}' \in \left(\mathcal{U}^x\right)^n$ and $b, b' \in \mathcal{U}^z$. If $\varphi(\bar{a};b) \leftrightarrow \varphi(\bar{a}';b)$ holds for every $\varphi(\bar{x},z) \in \Delta$ and $b \in B^z$, we write $\bar{a} \equiv_{\Delta,B} \bar{a}'$. We define $b \equiv_{A,\Delta} b'$ similarly.

In this notes, by a finitary type we mean a parameter-free type where only finitely many variables (among those displayed) occur. For every formula $\varphi(x_0, \dots, x_{n-1}; z) \in \Delta$ and

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every finitary type $\alpha(z) \subseteq L$ the language L^{p} contains the n-ary relation $r_{\varphi;\alpha}$. We abbreviate $r_{\varphi;\top}$ with r_φ. Indeed, the reader is invited to read $\alpha(z)$ as $\top$ untill Section 6, where it plays an essential role.

Below the symbol L^{p} will also denote the set of *atomic* formulas $r_{\varphi;\alpha}(x_0,\dots,x_{n-1})$ for any $x_0,\dots,x_{n-1} \in V$. Note that equalities are *not* included in L^{p}.

Given a pair of models $M \preceq N$, where M is ω-saturated and ω-homogeneous and N is $|M|^+$-saturated, we define a model $\mathcal{N}$ with domain N^x. We define the interpretation of $r_{\varphi;\alpha}$ in $\mathcal{N}$. If $\bar a = a_0,\dots,a_{n-1} \in N^x$, we declare $r_{\varphi;\alpha}(\bar a)$ true if

$$r_{\varphi;\alpha}(\bar a) \;\Leftrightarrow\; \alpha(M^z) \cap \varphi(\bar a; M^z) = \varnothing.$$

We say that $\varphi'(\bar x; z)$ is an alphabetic variant of the formula $\varphi(\bar x; z)$ if it is obtained by replacing the z-variables effectively occurring in $\varphi(\bar x; z)$ by other variables among those in z. Similarly, for finitary types. Clearly, if $\varphi'(\bar x; z)$ and $\alpha'(z)$ are alphabetic variants of $\varphi(\bar x; z)$ and $\alpha(z)$ respectively, then $r_{\varphi;\alpha}(\bar x)$ and $r_{\varphi';\alpha'}(\bar x)$ are equivalent.

The following is paramount.

1 Remark By the choice of Δ, the truth of $r_{\varphi;\alpha}(\bar a)$ only depends on the equivalence class of $\equiv_{\Delta,M}$ of the *components* of the tuple $\bar a$. Explicitely, if $\bar a = a_0,\dots,a_{n-1}$ and $\bar a' = a'_0,\dots,a'_{n-1}$ are such that $a_i \equiv_{\Delta,M} a'_i$ for every $i < n$, then

$$\#\qquad r_{\varphi;\alpha}(\bar a) \;\leftrightarrow\; r_{\varphi;\alpha}(\bar a').$$

By the remark, it would be natural to work in the quotient $\mathcal{N}/{\equiv_{\Delta,M}}$. But we prefer to keep the original domain $\mathcal{N}$ to have a common playground with L.

The set of maximally consistent set of formulas of the form $\varphi(x; b)$, where $\varphi(x; z) \in \Delta$ and $b \in M^z$ is denoted by $S_{x,\Delta}(M)$. We will often confuse it with either $\mathcal{U}^x/{\equiv_{\Delta,M}}$ or $\mathcal{N}/{\equiv_{\Delta,M}}$, with a preference for the latter.

A p-type is a set of formulas in $L^{\mathrm{p}}(\mathcal{N})$.

Note incidentally that the relation $x_0 \equiv_{\Delta,M} x_1$ is definable by a p-type. Namely the p-type containing the formulas $r_\vartheta(x_0, x_1)$ for every $\vartheta(x_0, x_1; z)$ of the form $\varphi(x_0; z) \leftrightarrow \varphi(x_1; z)$.

2 Fact Formulas in L^{p} are closed under adjunction of dummy variables, disjunction, conjunction, and existential quantification.

Proof. The claim about dummy variables is clear, so we tacitatly use it below.

Up to replacing $\varphi(\bar x; z)$ and $\alpha(z)$ with alphabetic variants, we can assume that the z-variables that effectively occur in $\varphi(\bar x; z)$ and $\alpha(z)$ are disjoint from those that occur in $\psi(\bar x; z)$ and $\beta(z)$. Therefore the following equivalences hold

$$\begin{aligned} r_{\varphi;\alpha}(\bar a) \vee r_{\psi;\beta}(\bar a) &\;\Leftrightarrow\; \forall z \in \alpha(M^z) \neg\varphi(\bar a; z) \;\vee\; \forall z \in \beta(M^z) \neg\psi(\bar a; z) \\ &\;\Leftrightarrow\; \forall z \in \alpha(M^z) \cap \beta(M^z) \;\; \neg\varphi(\bar a; z) \vee \neg\psi(\bar a; z) \end{aligned}$$

A similar argument applies to the conjunction of two formulas.

Finally, consider the formula $\exists y\, r_{\varphi;\alpha}(\bar{x},y)$. By the definition of Δ, we can assume that φ has the form

$$\varphi(\bar{x},y;z) \;=\; \bigvee_{i=1}^{m} \varphi_i(\bar{x},y;z)$$

where $\varphi_i(\bar{x},y;z) = \varphi'_i(\bar{x};z) \wedge \psi_i(y;z)$ and each $\varphi'_i(\bar{x};z)$ is a conjunction of formulas in $\bigcup_{x'\in V} L_{x';z}$ or negation thereof. Closure under existential quantification follows from the following equivalences

$$\begin{aligned}
\exists y\; r_{\varphi;\alpha}(\bar{a},y) &\;\Leftrightarrow\; \exists y \forall z\Big[z\in\alpha(M^z) \to \neg \bigvee_{i=1}^{m} \varphi_i(\bar{a},y;z)\Big] \\
&\;\Leftrightarrow\; \forall z\Big[z\in\alpha(M^z) \to \neg\Big(\bigvee_{i=1}^{m} \varphi'_i(\bar{a};z) \wedge \forall y\, \psi_i(y;z)\Big)\Big] \\
&\;\Leftrightarrow\; r_{\vartheta;\alpha}(\bar{a}),
\end{aligned}$$

where

$$\vartheta(\bar{x};z) \;=\; \bigvee_{i=1}^{m} \varphi'_i(\bar{x};z) \;\wedge\; \forall y\, \psi_i(y;z).$$

As $\exists y\,\psi_i(y;z) \in L_z \subseteq \Delta$, the above formula is in Δ as required. □

3 Fact (∞-p-saturation) Let $A \subseteq \mathcal{N}$ have arbitrary cardinality. Let $\bar{x}$ be a tuple of copies of x. Then every p-type $p(\bar{x}) \subseteq L^{\mathrm{p}}(A)$ that is finitely consistent in $\mathcal{N}$ is realized in $\mathcal{N}$.

Proof. Let $A' \subseteq \mathcal{N}$ be a small set containing a representative of every $\equiv_{\Delta,M}$ equivalence class that intersects A. Replacing the elements of A that occur in $p(\bar{x})$ by elements of A' we obtain an equivalent type $p'(\bar{x}) \subseteq L^{\mathrm{p}}(A')$. Note that every formula $r_{\varphi;\alpha}(\bar{x})$ there is a Δ-type $q(\bar{x}) \subseteq L(M)$ such that

$$\mathcal{N} \models r_{\varphi;\alpha}(\bar{a}) \;\Leftrightarrow\; N \models q(\bar{a})$$

Therefore the saturation of N implies that $p'(\bar{x})$ is realized in $\mathcal{N}$. □

2. Strong coheirs

We need a notion of independence that is stronger than the usual notion of coheir. The notion is adapted from that is [KNP]. We use parameter-free types (finitary types are assumed to be parameter-free) where [KNP] uses finitely many parameters from M. We beleave the difference is minor but it simplifies the exposition.

Let $p(z) \subseteq L(N)$. We say that $p(z) \subseteq L(N)$ is strongly finitely satisfiable in M if, for every finitary type $\alpha(z) \subseteq p$, and every formula $\varphi(z) \in p$, the type $\alpha(z) \wedge \varphi(z)$ is satisfied in M. Recall that a finitary type is one which is partameter-free and with finitely many variables.

Let $b \in \mathfrak{U}^z$ and $C \subseteq N$. When $\mathrm{tp}(b/C)$ is strongly finitely satisfiable in M, we write $b \perp_M C$. Explicitely, $b \perp_M C$ holds when $\alpha(M^z) \cap \varphi(M^z) \neq \varnothing$ for every $\varphi(z) \in L(C)$ and every finitary $\alpha(z)$ such that $b \models \alpha(z) \wedge \varphi(z)$.

4 Fact Every type $p(z) \subseteq L(N)$ that is strongly finitely satifiable in M has an extension to a complete type $q(z) \in S(N)$ that is strongly finitely satisfiable in M.

Proof. Let $q(z) \subseteq L(N)$ be maximal among the types that contain $p(z)$ and are strongly finitely satisfiable in M. We prove that $q(z)$ is complete. Suppose not, say it does not contain $\varphi(z)$ nor $\neg\varphi(z)$. Then there is a finitary type $\alpha(z) \subseteq p$ and a formula $\psi(z) \in p$ such that neither $\alpha(z) \wedge \psi(z) \wedge \varphi(z)$ nor $\alpha(z) \wedge \psi(z) \wedge \neg\varphi(z)$. This contradicts the strong finite satisfiability of $q(z)$. □

Let $\hat{m}$ be an enumeration of M and assume that $|z| = |\hat{m}|$. We define

$$\mathcal{B} \quad = \quad \left\{ b \in \mathfrak{U}^z \,:\, \hat{m} \equiv b \perp_M N \right\},$$

where $\equiv$ is the elementary equivalence in L (however, as $L_z \subseteq \Delta$, we could replace it with $\equiv_{\varnothing,\Delta}$). The N,Δ-topology on $\mathcal{B}$ is the one generated by the closed sets $\{b \in \mathcal{B} \,:\, \varphi(\bar{a}; b)\}$ for $\varphi(\bar{x}; z) \in \Delta$ and $\bar{a} \in N^{\bar{x}}$. The N,Δ-topology on $\mathcal{B}/{\equiv_{N,\Delta}}$ is the quotient of the topology on $\mathcal{B}$.

3. Morphism

A p-morphism is a map $f : \mathcal{N} \to \mathcal{N}$ such that for every $r_{\varphi;\alpha}(\bar{x}) \in L^{\mathrm{p}}$ and every $\bar{a} \in (\mathrm{dom} f)^{\bar{x}}$

$$\# \qquad \mathcal{N} \models r_{\varphi;\alpha}(\bar{a}) \;\Rightarrow\; \mathcal{N} \models r_{\varphi;\alpha}(f\bar{a}).$$

As $\equiv_{\Delta,M}$ is definable by a p-type, p-morphism induce well-defined maps on the quotient $\mathcal{N}/{\equiv_{\Delta,M}}$. We say that two p-morphism f and g are equivalent, and write $f \sim g$, if they induce the same map on $\mathcal{N}/{\equiv_{\Delta,M}}$. We often confuse equivalent p-morphism and we use the same symbol f to denote the map induced on the quotient.

An endomorphism is a total morphism. We write $\mathrm{End}(\mathcal{N})$ for the set of p-endomorphisms of $\mathcal{N}$.

5 Definition We endow $\mathrm{End}(\mathcal{N})$ with the topology that has as basic closed sets those of the form

$$\left\{ f \in \mathrm{End}(\mathcal{N}) \;:\; \mathcal{N} \models r_{\varphi;\alpha}\big(fa_1, \ldots, fa_n, c_1, \ldots, c_m\big) \right\}$$

for $\varphi(x; z) \in \Delta$ and $a_1, \ldots, a_n, c_1, \ldots, c_m \in N^x$.

Note that, by Remark 1, if $f \sim g$ then f and g belong to the same basic closed sets, therefore this topologies naturally induces a topology on $\mathrm{End}(\mathcal{N})/{\sim}$.

6 Fact With the above topology $\mathrm{End}(\mathcal{N})$ is T_1 (i.e. singletons are closed) and compact. Composition of p-morphism, which we denote by $*$, is a left continuous operation.

Proof. We prove that singletons are closed. In fact

$$\{f\} \quad = \quad \bigcap_{a\in N^x} \Big\{ g \in \mathrm{End}(\mathcal{N}) \,:\, a \equiv_{\Delta,M} ga \Big\}$$

therefore, as $\equiv_{\Delta,M}$ is defined by a p-type, $\{f\}$ is the intersection of closed sets.

We prove compactness. Assume the following intersection is finitely consistent

$$\#. \qquad \bigcap_{i<\lambda} \Big\{ f \in \mathrm{End}(\mathcal{N}) \,:\, \mathcal{N} \models r_{\varphi_i;\alpha_i}(f\bar{a},\bar{c}) \Big\}.$$

then the p-type

$$\bar{x} \equiv_{\Delta,M} \bar{a} \quad \cup \quad \{ r_{\varphi_i;\alpha_i}(\bar{x},\bar{c}) \,:\, i<\lambda \}$$

is finitely consistent in $\mathcal{N}$. By ∞-p-saturation, it is realized by some $\bar{a}'$. By Fact 7 there is a p-endomorphism $\mathcal{N}$ that maps $\bar{a}$ to $\bar{a}'$. This proves that the set in # is non empty.

We prove left continuity of composition. That is for any given $f \in \mathrm{End}(\mathcal{N})$ we prove that the function $\tau_f : \mathrm{End}(\mathcal{N}) \to \mathrm{End}(\mathcal{N})$ that maps $g \mapsto g * f$ is continuous. Pick any basic closed set $\{h : r_{\varphi\alpha}(h\bar{a},\bar{c})\}$. Then

$$\tau_f^{-1}\big[\{h : r_{\varphi;\alpha}(h\bar{a},\bar{c})\}\big] \quad = \quad \{g \,:\, r_{\varphi;\alpha}(g(f\bar{a}),\bar{c})\}$$

is closed, as required. □

If the converse of the implication in # also holds, f is called a partial p-isomorphism. For the back-and-forth argument in the following proof we need to take inverse of partial p-isomorphisms. Strictly speaking partial p-isomorphism need not be invertible but for our puposes it suffices to have a p-morphism g such that $f\circ g$ and $g\circ f$ are equivalent to the identity on range$f$, respectively dom$f$.

7 Fact Every p-morphism extends to a p-morphism defined on the whole of $\mathcal{N}$. Also, every p-isomorphism extends to a p-automorphism of $\mathcal{N}$.

Proof. Let $f : \mathcal{N} \to \mathcal{N}$ be a p-morphism. Let $\bar{a}$ be an enumeration of dom$f$ and let $b \in N^x$. Let $p(\bar{x}) = \text{p-tp}(\bar{a})$ and let $q(\bar{x},y) = \text{p-tp}(\bar{a},b)$. By ∞-p-saturation $\exists y\, q(\bar{x},y)$ is equivalent to the type containing the existential closure of the formulas in $q(\bar{x},y)$. By Fact 2, the latter is equivalent to a p-type, which is necessarily coicides with $p(\bar{x})$. Then $q(f\bar{a},y)$ is consistent hence by ∞-p-saturation it is realized by some $c \in N^x$. Then $f \cup \{\langle b,c\rangle\}$ is a p-morphism defined in b that extends f.

The second claim is obtained by back-and-forth. □

4. Duality

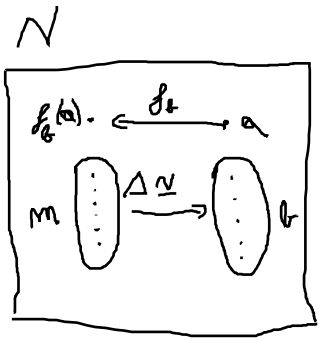
Recall that $\hat{m}$ is an enumeration of M and $|z| = |\hat{m}|$. For convenient notation, x is assumed to be of infinite length – then, up to equivalence, all formulas in $L_z(\mathfrak{U})$ have the form $\varphi(a;z)$ for some $\varphi(x;z) \in L$ and $a \in \mathfrak{U}^x$.

8 Remark We stress a simple fact which is used in the proof below. Recall that $\varphi'(a;z)$ and $\alpha'(z)$ are alphabetic variants of $\varphi(a;z)$ and $\alpha(z)$ if they are obtained by replacing the variables effectively occurring in $\varphi(a;z)$ and $\alpha(z)$ by other variables among those in z. The following are clearly equivalent

1. $\alpha(M^z) \cap \varphi(a;M^z) = \varnothing$
2. $\hat{m} \not\models \alpha'(z) \wedge \varphi'(a;z)$ for every alphabetic variants of $\varphi(a;z)$ and $\alpha(z)$.

9 Fact There is a continuous bijection $\mathcal{B}/\equiv_{N,\Delta} \to \mathrm{End}(\mathcal{N})/\sim$.

Proof. In 1 below, we define a map $b \mapsto f_b$ from $\mathcal{B}$ into $\mathrm{End}(\mathcal{N})$. In 2, we define a map $f \mapsto b_f$ from $\mathrm{End}(\mathcal{N})$ into $\mathcal{B}$. The reader may check that these maps induce maps between $\mathcal{B}/\equiv_{N,\Delta}$ and $\mathrm{End}(\mathcal{N})/\sim$ that are each other's inverse. In 3 we prove the continuity of the first map.



1. Let $b \in \mathcal{B}$ be given. We define

$$p(x,y,z) \quad = \quad \{\varphi(y;z) \to \varphi(x;\hat{m}) \,:\, \varphi(x;z) \in \Delta\}.$$

We claim that if $a \equiv_{\Delta,M} a'$ then $p(x,a,b) = p(x,a,b)$. In fact, $\varphi(a;b) \nleftrightarrow \varphi(a';b)$ would contradict $b \perp_M N$. Also, note that $p(x,a,b)$ is consistent, otherwise $\neg\exists x \bigwedge_i \varphi_i(x;\hat{m})$, for some finitely many formulas $\varphi_i(x;z) \in \Delta$ such that $\varphi_i(a;b)$. But this would contradict $b \equiv \hat{m}$.

Therefore it is possible to define $f_b(a)$ to be some/any realization of $p(x,a,b)$. We prove that f_b is a p-morphism.

Let $\psi(x_1,\dots,x_n;z) \in \Delta$, $\alpha(z) \subseteq L$, and $a_1,\dots,a_n \in \mathfrak{U}^x$ be such that $\neg r_{\psi;\alpha}(f_b a_1,\dots,f_b a_n)$. Then $\alpha(M^z) \cap \psi(f_b a_1,\dots,f_b a_n;M^z) \neq \varnothing$. Pick some alphabetic variants such $\alpha'(\hat{m})$ and $\psi'(f_b a_1,\dots,f_b a_n;\hat{m})$. From the definition of f_b we obtain $\psi'(a_1,\dots,a_n;b)$ and, as $b \equiv \hat{m}$, we also have $\alpha'(b)$. Finally, as $b \perp_M N$, we conclude that $\alpha'(M^z) \cap \psi'(a_1,\dots,a_n;M^z) \neq \varnothing$, i.e. that $\neg r_{\psi;\alpha}(a_1,\dots,a_n)$.

2. Let $f : \mathcal{N} \to \mathcal{N}$ be an p-epimorphism. Let $p(z)$ be the type containing the formulas $\varphi(a;z)$ for $\varphi(x;z) \in \Delta$ and $a \in N^x$ such that $\varphi(fa;\hat{m})$. Clearly $p(z) \vdash \hat{m} \equiv z$ and is complete over N,Δ. We prove that $p(z)$ is strongly finitely satisfied in M, then b_f is some/any realization of $p(z)$.

Suppose for a contradiction $p(z)$ is not strongly finitely satisfied in M. Then $\alpha(M^z) \cap \psi(a_1,\dots,a_n;M^z) = \varnothing$, where $\alpha(z) \subseteq p$ is a finitary type and

$$\psi(x_1,\dots,x_n;z) \quad = \quad \bigwedge_{i=1}^{n} \varphi_i(x_i;z)$$

for some $\varphi_i(x;z) \in L$ and $a_i \in N^x$ such that $\varphi_i(fa_i;\hat{m})$ for every i. Then $r_{\psi;\alpha}(a_1,\dots,a_n)$ and $\neg r_{\psi;\alpha}(fa_1,\dots,fa_n)$. This is not possible because f is a p-morphism.

3. We prove that the map defined in 1 is continuous. Pick any basic closed set in $\mathrm{End}(\mathcal{N})$, say $\{h : r(h\bar{a},\bar{c})\}$. We claim that the set $\{b : r_{\psi;\alpha}(f_b\bar{a},\bar{c})\}$ is closed. Indeed, by the definition of f_b in 1,

$$\{b : r_{\psi;\alpha}(f_b\bar{a},\bar{c})\} \;=\; \Big\{b : b \models \exists \bar{x}\Big[\bigwedge_{i=1}^{n} p(x_i,a_i,z) \wedge r_{\psi;\alpha}(\bar{x},\bar{c})\Big]\Big\}$$

is closed in $\mathcal{B}$ because $r_{\psi;\alpha}(\bar{x},\bar{c})$ is definable by a N,Δ-type. □

5. Semigroups

Assume $\mathcal{U}$ is a unital semigroup (a monoid) w.r.t. some definable operation which we denote by $\cdot$. The unit is denoted by 1. In this section we could work with $|x| = |z| = 1$ but, for uniformity with the previous section we take $|x| = |z| = \omega$. We write $z \cdot x$ for coordinatewise multiplication. In this section $\Delta(x;z)$ contains the formulas $\varphi(z \cdot x)$ for all $\varphi(x) \in L$.

10 Fact Assume that $a \cdot M = M$ for all $a \in M$, which holds in particular when $\mathcal{U}$ is a group. For $b \in \mathcal{B}$, let $g_b : \mathcal{N} \to \mathcal{N}$ be the function that maps $a \mapsto b \cdot a$. Then $b \mapsto g_b$ induces a bijection between $\mathcal{B}/\equiv_{N,\Delta}$ and $\mathrm{End}(\mathcal{N})/\sim$.

Proof. In 1 below we prove that g_b is a p-morphism. In 2 we define a map $g \mapsto b_g$ from $\mathrm{End}(\mathcal{N})$ into $\mathcal{B}$. The reader may check that these maps induce maps between $\mathcal{B}/\equiv_{N,\Delta}$ and $\mathrm{End}(\mathcal{N})/\sim$ that are each other's inverse. In 3 we prove the continuity of the map $b \mapsto g_b$.

1. Assume $r_{\psi;\alpha}(a_1,\ldots,a_n)$. We prove $r_{\psi;\alpha}(g_b a_1,\ldots,g_b a_n)$. Suppose not, then $\alpha(M^z) \cap \psi(M^z \cdot b \cdot a_1,\ldots;M^z \cdot b \cdot a_n) \neq \varnothing$. From $b \perp_M N$, we obtain $\alpha(M^z) \cap \psi(M^z \cdot a_1,\ldots;M^z \cdot a_n) \neq \varnothing$. This contradicts $r_{\psi;\alpha}(a_1,\ldots,a_n)$.

2. Let $g : \mathcal{N} \to \mathcal{N}$ be a p-morphism. Let $p(y)$ be the type containing $\varphi(\hat{m} \cdot y \cdot a)$ for all $\varphi(x) \in L$ and $a \in N^x$ such that $\varphi(\hat{m} \cdot ga)$. We prove that $p(y)$ is strongly finitely satisfied in M. If for a contradiction $\alpha(M^z) \cap \psi(a_1,\ldots,a_n;\hat{m} \cdot M^z) = \varnothing$, where

$$\psi(x_1,\ldots,x_n;z) \;=\; \bigwedge_{i=1}^{n} \varphi_i(z \cdot x_i)$$

for some $\varphi_i(x) \in p$ and $a_i \in N^x$ such that $\varphi_i(\hat{m} \cdot ga_i)$ for every i. As $\hat{m} \cdot M^z = M^z$, we have that $r_\psi(a_1,\ldots,a_n)$ and $\neg r_\psi(ga_1,\ldots,ga_n)$. This is not possible because g is a p-morphism. Then any b_g that realizes $p(z)$ is such that $g \sim g_{b_g}$.

3. We prove that the map defined in 1 is continuous. Pick any basic closed set in $\mathrm{End}(\mathcal{N})$, say $\{h : r(h\bar{a},\bar{c})\}$ where $\bar{a} = a_1,\ldots,a_n$ and $\bar{c} = c_1,\ldots,c_m$. We claim that the set $\{b : r(g_b\bar{a},\bar{c})\}$ is closed. Indeed, by the definition of g_b in 1,

$$\{b : r(g_b\bar{a},\bar{c})\} \;=\; \{b : r(b \cdot a_1,\ldots,b \cdot a_n,\bar{c})\}$$

is closed in $\mathcal{B}$ because $r(w \cdot a_1,\ldots,w \cdot a_n,\bar{c})$ is definable by a N,Δ-type $p(w,\bar{a},\bar{c})$. □

6. Ellis's semigroup

We write $\mathrm{Aut}(N/\{M\})$ for the group of automorphisms of N that fix M setwise. In this section we show that $\mathrm{End}(\mathcal{N})$, endowed with a topology that is finer than that introduced

above, is homeomorphic to the Ellis semigroup associated to the action $\mathrm{Aut}(N/\{M\})$ on N^x. Note that this is essentially equivalent to the action of $\mathrm{Aut}(M)$ on $S_{x,\Delta}(M)$ – we chose the former based on personal preference.

We recall the definition of this semigroup, which we denote by $\mathrm{El}_x(N,M)$. On $\mathcal{N}$ we assume the topology generated by the closed sets of the form $\varphi(N^x;b)$ for $\varphi(x;z)\in\Delta$ and $b\in M^z$. This we call the logic topology. We identify $\mathrm{Aut}(N/\{M\})$ with a subset of $(N^x)^{N^x}$, the set of all maps from N^x into itself, which we endow with the product topology. The Ellis semigroup is the closure of $\mathrm{Aut}(N/\{M\})$ in the product topology. The following theorem proves that $\mathrm{El}_x(N,M)$ concides with $\mathrm{End}(\mathcal{N})$.

11 Theorem Assume M is λ-homogeneous (where $\lambda=|z|$). Then $\mathrm{End}(\mathcal{N})$ is a closed subset of $(N^x)^{N^x}$, and $\mathrm{Aut}(N/\{M\})$ is dense subset of it.

Note that the topology that $\mathrm{End}(\mathcal{N})$ inherits from $(N^x)^{N^x}$ is finer than the topology introduced in Definition 5.

Proof. It is clear that $\mathrm{End}(\mathcal{N})$ is a closed set containing $\mathrm{Aut}(N/\{M\})$. Let $f\in\mathrm{End}(\mathcal{N})$ and $a_1,\dots,a_n\in N^x$ be given. Let $\varphi_i(x;z)\in\Delta$ and $b\in M^z$ be such that $fa_i\models\varphi_i(x;b)$. We need to prove that there is $g\in\mathrm{Aut}(N/\{M\})$ such that $ga_i\models\varphi_i(x;b)$. Let $\alpha(z)=\mathrm{tp}(b)$. Let

$$\psi(x_1,\dots,x_n;z) \;=\; \neg\bigwedge_{i=1}^{n}\varphi_i(x_i;z)$$

As $\neg r_{\psi;\alpha}(fa_1,\dots,fa_n)$ and f is a p-morphism, $\neg r_{\psi;\alpha}(a_1,\dots,a_n)$. Then there is $c\models\alpha(z)$ such that $\varphi_i(a_i;c)$ for every i. As M is λ-homogeneous there is a $g\in\mathrm{Aut}(N/\{M\})$ such that $gc=b$, then $\varphi_i(ga_i;b)$ as requiured. □

7. Cores

A p-retraction of $\mathcal{N}$ onto $\mathcal{A}$ is a total p-morphism $f:\mathcal{N}\to\mathcal{A}$ that is (equivalent to) the identity on $\mathcal{A}$. Retractions are exactly the idempotent elements of $\mathrm{End}(\mathcal{N})$. In this section we characterize the minimal idempotents of $\mathrm{End}(\mathcal{N})$, that is, the p-morphisms $u\in\mathrm{End}(\mathcal{N})$ such that $\mathrm{End}(\mathcal{N})*u$ is a minimal left ideal. These are the retractions onto substructures of $\mathcal{N}$ named *cores*.

A set $A\subseteq\mathcal{N}$ is p-maximal if it satisfies the following (obviously) equivalent conditions for every $\bar a\in A^{<\omega}$ and every p-morphism f defined on $\bar a$:

a. $r_\varphi(\bar a)\;\Leftrightarrow\;r_\varphi(f\bar a)$ for every $r_\varphi\in L^{\mathrm{p}}$

b. $\text{p-tp}(\bar a)\;=\;\text{p-tp}(f\bar a)$.

12 Fact The following are equivalent

1. $A \subseteq \mathcal{N}$ is p-maximal
2. $p(\bar{x}) = \text{p-tp}(\bar{a})$ is a maximally consistent p-type for every $\bar{a} \in A^{\bar{x}}$ and every finite $\bar{x}$
3. $p(\bar{y}) = \text{p-tp}(\bar{c})$ is a maximally consistent p-type for $\bar{c}$ any enumeration of A
4. every p-morphism defined on A is a p-isomorphism between A and its image.

Proof. 1⇒2. Negate 2. Suppose $\text{p-tp}(\bar{a})$ is not maximal for some finite $\bar{a} \in A^{\bar{x}}$. Pick some $\bar{a}' \in A^{\bar{x}}$ such that $\text{p-tp}(\bar{a}) \subset \text{p-tp}(\bar{a}')$. Then $f = \{\langle \bar{a}, \bar{a}' \rangle\}$ is a p-morphism that contradicts the p-maximality of A.

2⇒3. If $\bar{c}$ is an enumeration of A that witnesses the failure of 3, then some finite subtuple of $\bar{c}$ witnesses the failure of 2.

3⇒4. Negate 4. Let f be a p-morphism defined on A that is not a p-isomorphism. Let $\bar{a} \in A^{\bar{x}}$ be such that $\text{p-tp}(\bar{a}) \smallsetminus \text{p-tp}(f\bar{a}) \neq \varnothing$. To contradict 3, it suffices to extend f to a p-morphism defined on A. This is possible by ∞-p-homogeneity.

4⇒1. Clear. □

13 Example Assume that $a \in N^x$ is such that $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}}\varnothing$ for every $\varphi(x;z) \in L$. Then the orbit of a under $\text{Aut}(\mathfrak{U}/\{M\})$ is p-maximal.

Proof. First consider the case when $\varphi(a; M^z)$ is definable over $\varnothing$. We prove that $\{a\}$ is p-maximal.

Let $\varphi'(z) \in L$ be such that $\varphi(a; M^z) = \varphi'(M^z)$. Then, if we set $\varphi''(x;z) = \varphi(x;z) \leftrightarrow \varphi'(z)$, the formula $r_{\varphi''}(a)$ holds.

If f is a p-morphism then $r_{\varphi''}(fa)$ also holds. Then $a \equiv_{\Delta,M} fa$ follows. This shows that p-morphism are trivial, i.e. equivalent to the identity, and therefore p-isomorphisms.

Now assume that $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}}\varnothing$.????????? □

A set $\mathcal{J} \subseteq \mathcal{N}$ is called a core if it is p-maximal and, among the p-maximal sets, it is maximal w.r.t. inclusion.

14 Fact Let $\mathcal{J}$ be a core. Let $\mathcal{A}$ be a p-maximal set. Then every p-morphism $f : \mathcal{A} \to \mathcal{J}$ extends to a p-isomorphism between $\mathcal{A}$ and $\mathcal{J}$.

Proof. Reason as in the proof of Fact 7 with the same notation but choose $b \in \mathcal{A}$. Let f' be a p-morphism defined on $\mathcal{A}$ that extends $f \cup \{\langle b, c \rangle\}$. Then $f'[\mathcal{A}]$, i.e. the image of $\mathcal{A}$ under f', is a p-maximal set that contains $\mathcal{J}$. By maximality $f'[\mathcal{A}]$ coincides with $\mathcal{J}$, hence $c \in \mathcal{J}$. □

By back-and-forth we obtain the following.

15 Corollary (∞-p-homogeneity of the core) If $\mathcal{J}$ and $\mathcal{J}'$ are cores, then every p-morphism $f : \mathcal{J} \to \mathcal{J}'$ extends to a p-isomorphism between $\mathcal{J}$ and $\mathcal{J}'$.

A p-retraction of $\mathcal{N}$ onto $\mathcal{A}$ is a total p-morphism $f : \mathcal{N} \to \mathcal{A}$ that is equivalent to the identity on $\mathcal{A}$.

16 Theorem If $\mathcal{J}$ and is a core, then there is a p-retraction of $\mathcal{N}$ onto $\mathcal{J}$.

Proof. Let $\bar{c}$ enumerate $\mathcal{N}$. Let $\bar{a}$ be a tuple such that p-tp$(\bar{a}) \supseteq$ p-tp$(\bar{c})$ is p-maximal. Let $f = \{\langle \bar{c}, \bar{a} \rangle\}$. Then $f : \mathcal{N} \to \mathcal{A}$ is a p-morphism onto the p-maximal set $\mathcal{A} =$ range($\bar{a}$). Then $f \restriction \mathcal{J}$ is a p-isomorphism from $\mathcal{J}$ into $\mathcal{A}$. As $\mathcal{J}$ is a core, $f \restriction \mathcal{J}$ is p-isomorphism between $\mathcal{J}$ and $\mathcal{A}$. Then $f \circ (f \restriction \mathcal{J})^{-1}$ is a retraction of $\mathcal{N}$ onto $\mathcal{J}$. □

17 Fact The following are equivalent

1. $a \in \bigcap \{\mathcal{J} : \mathcal{J} \text{ is a core}\}$
2. $\varphi(a; M^z)$ is definable over $\text{acl}^{\text{eq}} \varnothing$, for every formula $\varphi(x; z) \in L$.

Proof. ??? □

18 Fact $\mathcal{J}$ is compact subset of $\mathcal{N}$.

Proof. ??? □

8. Esperimenti

Recall that $\hat{m}$ is an enumeration of M and $|z| = |\hat{m}|$.

The following fact is well-known. It entails that the relation a and b are at distance ≤ 1 in the Lascar graph is type-definable (this is the relation defined by 1 in the fact).

19 Fact The following are equivalent

1. the $a \equiv_{M'} b$ for some model M'
2. $a \equiv_{m'} b$ for some $m' \equiv \hat{m}$.

Proof. 2⇒1. Clear.

1⇒2. Negate 2. Then $\alpha(z) \vdash \varphi(a; z) \nleftrightarrow \varphi(b; z)$ for some $\varphi(x; z) \in L$ and some $\alpha(z) \in \text{tp}(\hat{m})$. As every model witnesses $\alpha(z)$, this negates 1. □

Let $l_1(x, x') \subseteq L$ be the type that says that x, x' are at distance ≤ 1 in the Lascar graph.

20 Fact There is a p-type defining $l_1(x, x')$.

Proof. Consider the p-type containing the formulas $r_{\vartheta,\alpha}(x_0, x_1)$ for every $\vartheta(x_0, x_1; z)$ of the form $\varphi(x_0; z) \nleftrightarrow \varphi(x_1; z)$, where $\alpha(z)$ is the type $\mathrm{tp}(\hat{m})$ restricted to the z-variables that effectively occur in $\varphi(x; z) \in \Delta$. □

21 Fact If $\neg l_1(a, b)$, then $f a \not\equiv_M f b$, for every p-morphism f.

Proof. Assume $\neg l_1(a, b)$. Then $\alpha(z) \vdash \varphi(a; z) \nleftrightarrow \varphi(b; z)$ for some $\varphi(x; z) \in L$ and some $\alpha(z) \in \mathrm{tp}(\hat{m})$. Then $r_{\vartheta,\alpha}(a, b)$, where $\vartheta(x_0, x_1; z)$ is the formula $\varphi(x_0; z) \nleftrightarrow \varphi(x_1; z)$. Then $r_{\vartheta,\alpha}(f a, f b)$ also holds. As $\alpha(z)$ is satisfable in M, we obtain $f a \not\equiv_M f b$. □

22 Corollary If $f : N \to \mathcal{J}$ is a retraction, then $l_1(a, f a)$ for every $a, b \in \mathcal{N}$.

9. Notes and references

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